

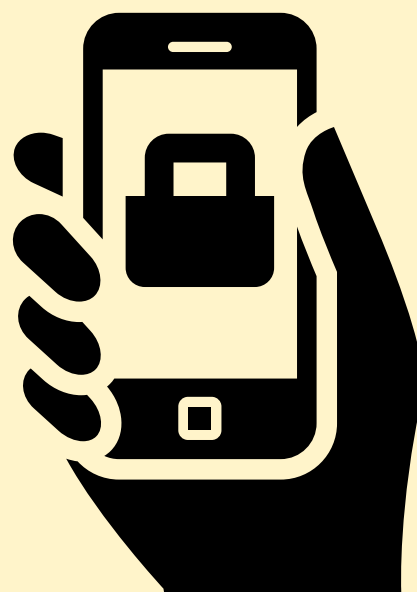
can be eased a long way if you took some of the following measures:

- **Social = unsafe:** We all know that almost all popular online services and software, including social media websites, are on the NSA's radar. Not just abuse of privacy, you should be cautious about PII (personally identifiable information) being compromised by scamsters, malware players as well – often leading to potential damages. Leaving a trail of your activity on Facebook, Twitter, Google+ or any other social hotspot is only as good as leaving yourself vulnerable and exposed. Not a good idea.
- **Encrypted OS:** If you can't encrypt your OS (Windows, Mac, or Linux), you simply aren't trying hard enough. TrueCrypt, as mentioned above is a great tool to secure files and folders on your system, at an OS level. You can also check out Tails (<https://tails.boum.org>), a live OS that can boot off a USB stick, leaves no trace on the computer you are using and deploys state-of-the-art cryptographic tools to encrypt your files, emails and instant messaging. Similarly, BackTrack Linux (<http://www.backtrack-linux.org>) is also very secure. For supported Android smartphones, rooting and installing Replicant (<https://www.replicant.us>), F-Droid (<https://f-droid.org>), and The Guardian Project (<https://guardianproject.info>) are good secure anonymizing options available to users currently.
- **Email? ProtonMail:** At this level of paranoia, no other email service will cut it for you, only the "best" will do. Developed by MIT and CERN scientists, with servers housed in Switzerland (world's most neutral spot), this email service has even the NSA sweating it out. It's currently open to access by invitation only, and in beta phase, as the scientists who made it are also actively researching in math, computing and physics. But if you can manage to get access to ProtonMail (<https://protonmail.ch>), which also has supported mobile apps, you can say goodbye to any other non-secure email service you've been using all this while. Alternatively, use disposable email services – GuerillaMail (<https://www.guerrillamail.com>) is one of the best in the business.

FOR YOUR PHONE

From the outset, it's difficult to anonymize one's activity over smartphones (because of all the radio communication carried out by them). Still, preventable measures to reduce your exposure to the bare minimum, is possible. Use Replicant (as mentioned elsewhere) to boot an alternate OS onto supported Android ROMs. Use CryptoSMS, for encrypted SMS (<https://www.cryptosms.org>) between devices. The Orbot app is essentially Tor for Android (<https://guardianproject.info/apps/orbot>) with Orweb, being a Tor-like browser. Gwibberbot is a secure chat service: (<https://guardianproject.info/apps/gibber>), while the Linphone is a secure Skype-like VOIP client: (<https://linphone.org>) similar to RedPhone. For encrypted phone calls, use Ostel (<https://guardianproject.info/apps/ostel>) and make sure you use Pul Wi-Fi scanner (<http://pulwifi.en.uptodown.com/android>) to check if your Wi-Fi password and network is secure.

The Guardian Project (<https://guardianproject.info>) and F-Droid (<https://f-droid.org/about>) has a whole host of software and apps for smartphones to ensure greater privacy and anonymity online through your smart devices (tablets, too). The Tor Project endorses the following apps from F-Droid: Xabber, Offline Calendar, Plumble, K-9 Mail and APG, OSMAnd-, Launch App Ops, Permissions, CatLog, OS Monitor, and Intent Intercept, among other things. For the full list, and steps to make a Google Nexus 7 tablet completely bug-free, go here: <https://blog.torproject.org/blog/mission-impossible-hardening-android-security-and-privacy#Software>



- **OTR:** A protocol called Off-the-record, which is a champion of end-to-end encrypted communication, is emerging as another blind spot as far as the NSA's concerned and resonating among privacy activists online. Using services based on OTR protocol are critical to have the maximum chance of ensuring the privacy of your communication. Check out Cryptocat (<https://crypto.cat>) and BitlBee (<https://www.bitlbee.org>).
- **Leave no trace:** Apart from encryption to anonymize yourself while you're in the system, you must not forget to wipe clean your tracks. This is especially true if you're accessing personal files in a non-secure location (like a work computer or cyber cafe). For this, we recommend simply BleachBit (<http://bleachbit.sourceforge.net>), which single-handedly frees disk space and tirelessly guards your privacy. Free cache, delete cookies, clear Internet history, shred temporary files, delete logs,

and discard junk you didn't know was there. Beyond simply deleting files, BleachBit includes advanced features such as shredding files to prevent recovery, wiping free disk space to hide traces of files deleted by other applications, too.

- **Secret Sauce:** Since you're trying to keep the NSA's nose out of your online activities, here's a documented "catastrophic" combination of tools when deployed together will as good as guarantee your privacy online. According to freshly leaked NSA documents from December 2014, if users incorporated a variety of security measures, such as using Tor (<https://torproject.org>), to connect online, CSspace (<http://www.anonymous-p2p.org/cspace.html>), to send instant messages online, transfer files and ZRTP (an encryption technique similar to OTR) to make VOIP calls, the combination made users "nearly invisible to the NSA," according to a security report. 